

한정연

JungYun Han

AI Engineer / LLM, RAG, Agent, NLP Modeling

At Samsung SDS I built and operated generative AI quality analytics, RAG and agent based automation, news retrieval and recommendation, and logistics risk NLP models.



CAREER

Samsung SDS, Senior Data Consultant (Oct 2022 –)
Institute for Basic Science (IBS), Research Associate (Mar 2018 – Aug 2022)

RESEARCH

Ph.D. in Theoretical Physics, four SCI journal papers
Quantum computing patent filed (KR 2025-0139679)

CERTIFICATE

Engineer Big Data Analysis, Engineer Information Processing, OPlc IH, OPI IH

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Four results, each with a number behind it

4→8

Monthly deliveries

RAG, agents, Validator gate

90%

Intent accuracy

Fine-tuned Longformer

4.0+

Satisfaction (out of 5)

Contributed to a KRW 3B follow-on

24h→2h

Risk response lead time

BERT detection, quarterly retraining

01 RAG / AGENT

Exam authoring automation

Agents retrieve past items and rubric documents, then draft questions, answer code, and review criteria.

02 LLM EVALUATION

GenAI quality analytics

I classified intent from question and answer pairs, and verified label quality before use.

03 NLP OPERATIONS

Search and risk operations

I kept quality up in production with embedding retrieval, reranking, threshold tuning, and retraining.

What the role needs, and where I have done it

01 Building GenAI systems end to end

I planned and built RAG and agent based exam authoring.
Monthly deliveries rose from 4 to 8; lead time fell 50%.

02 LLM evaluation and verification

I checked 3-LLM majority labels against 7,000 human labels.
Unanimous cases ran above 95% accuracy; intent classification hit 90%.

03 Agent architecture

I split generation and validation into separate LangGraph agents.
A Validator gate checks code execution, reproducibility, and data leakage.

04 Prompt engineering

I designed labeling prompts and per-department persona prompts, which held up label quality and recommendation relevance.

05 Operating AI services

I ran threshold tuning and quarterly hard-negative retraining.
Risk response time fell from 24 hours to 2.

06 Deployment under constraints

I operated Docker, GitLab CI, and cron batches on Samsung Cloud Platform and in air-gapped on-premise networks.

Agentic Exam Automation

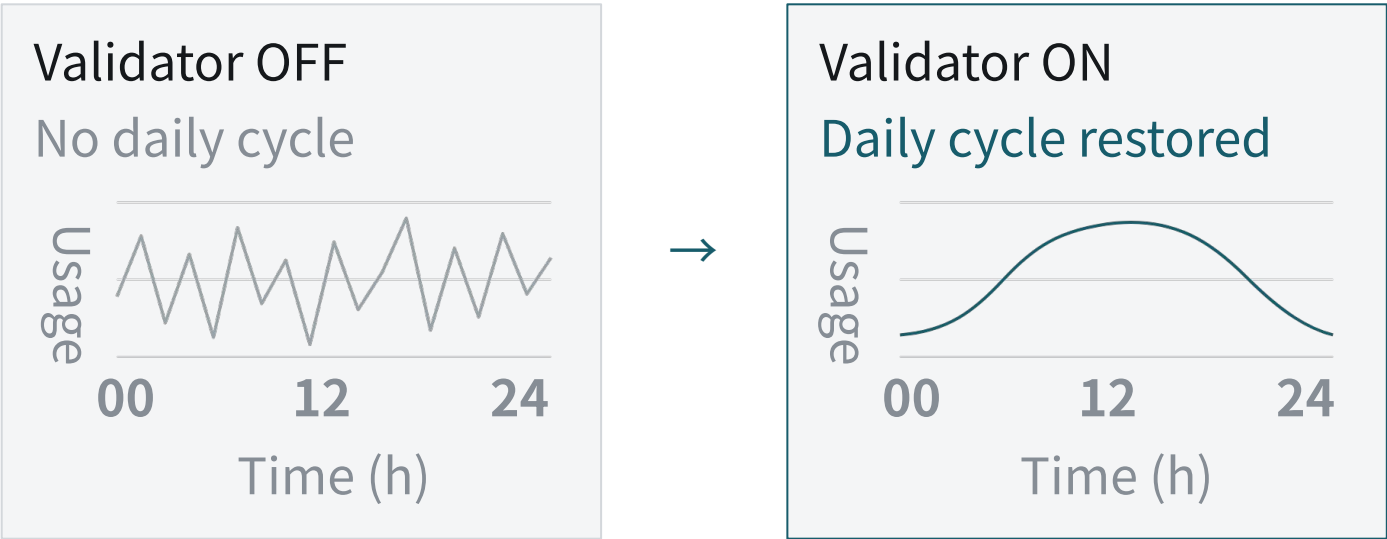
4 → 8

Monthly exam deliveries

Context

Exam production for Samsung Electronics ran on hand labor, from dataset creation to questions, answer code, and translation. Deadlines and quality review were the bottleneck.

Synthetic data check: hourly electricity usage



Automation and quality control

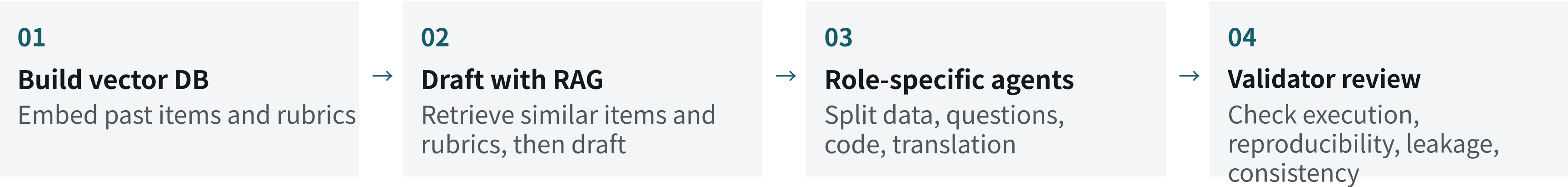
I organized past items and rubrics at the item level into a vector database.

I split the work into role-specific agents, with generation and validation as separate stages.

The Validator checks code execution, reproducibility, and leakage, plus synthetic-data distribution and periodicity.

I documented the criteria so reviewers share one checklist. Deliveries rose 4 to 8, lead time fell 50%. Validation rules mattered more than generation speed.

Workflow with generation and validation split

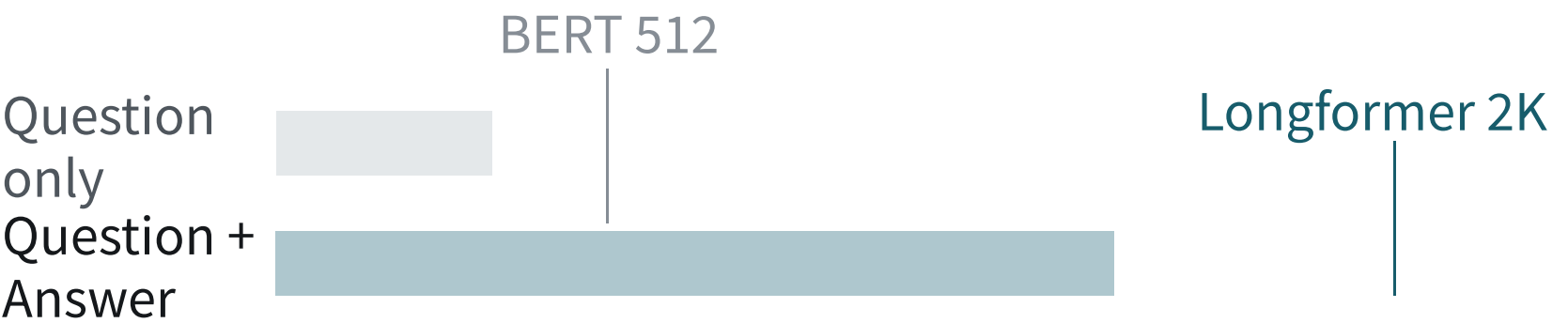


Feedback Classification

Context

During the beta of GAUSS, Samsung's generative AI, we collected NPS on user feedback but could not tell which question types scored poorly. I built a model that classifies feedback into 20 intent types, then read NPS by type to find what to fix.

Token length vs. limits



Analysis and model design

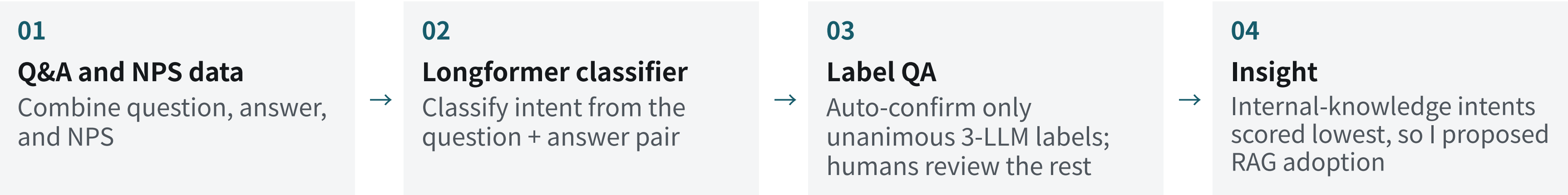
To see differences by intent rather than one average, I paired multi-turn questions with model answers as a single input.

Many inputs ran past 512 tokens, so I fine-tuned a 2K-token Longformer instead of BERT.

Three LLMs labeled each case independently. Unanimous cases auto-confirmed; disagreements went to human review.

Checked against about 7,000 existing human labels, unanimous cases exceeded 95% accuracy, with a 95% confidence interval within $\pm 1\%$.

Feedback analysis and verification flow



90%
Longformer accuracy

9→3MM
Labeling & verification effort

News Recommendation

4.0+

W3B

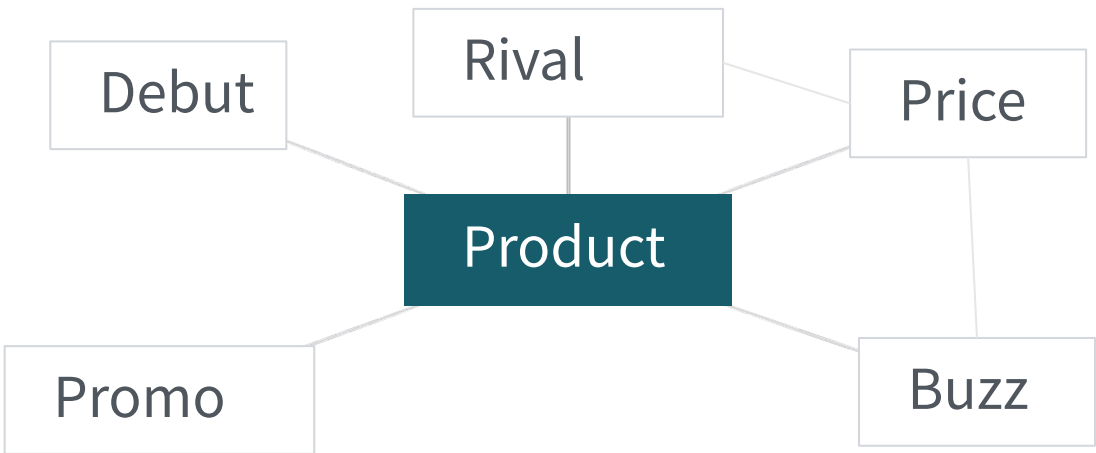
User satisfaction (out of 5)

Follow-on engagement

Context

Each Samsung Electronics department tracked different issues, and staff picked news by hand. No structured data described the departments, so the recommendation criteria had to be defined first.

Knowledge graph from TF-IDF co-occurrence



Design and contribution

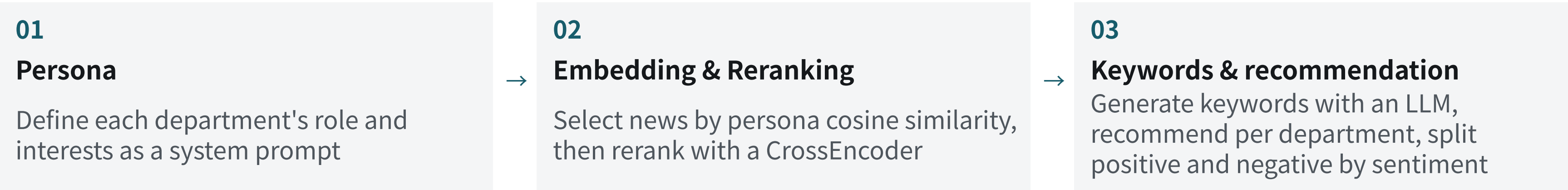
I interviewed business staff to define each department's role, products, competitors, and interests as a persona.

Cosine similarity pulled a wide candidate set; a CrossEncoder reranked it above keyword matching.

A TF-IDF co-occurrence graph showed the context of new keywords, and BERT sentiment split opportunities from risks.

The work that mattered most was defining the inputs with the business, before any modeling.

Recommendation flow



Logistics Risk Detection

24h → 2h

Risk response lead time

Context

Logistics operators watched news around the clock by hand. The daily article volume stretched response times.

Same strike keyword, different risk: why BERT

Port strike drags on; shipment delays feared	Risk High
Union and management in talks to end the strike	Low

Judgment and execution

- A missed risk cost more than a false alarm, so recall came first, with thresholds tuned to the alert volume operators could handle.
- Keywords could not separate subject and timing, so XGBoost narrowed candidates and BERT read context to classify severity.
- False alarms that came with higher recall fed hard-negative mining and quarterly retraining.

Pipeline

1,000 daily articles narrow to 3–5 alerts; response time fell from 24 hours to 2.



1,000

Crawl logistics news into Hive

Collect and clean

< 200

XGBoost
Recall-first risk filter
Risk-keyword similarity features

< 100

Condense
BART-CNN summaries
Cluster out duplicates

3–5

Auto alert, 2-hour response
BERT severity classification

ShockFlow AI: Event-Feature Demand Forecasting

I joined NYC event data with bike-share demand to test whether external features help, then fed the same forecasts into rebalancing and dynamic pricing.

1. Data collection

- 20.34M trips
- 63,070 permitted events
- News (GDELT), weather (NOAA)
- GBFS live inventory

2. LLM structuring

Extract type, severity, direction from news; schema checks and publish times block leakage

3. Graph / vector DB

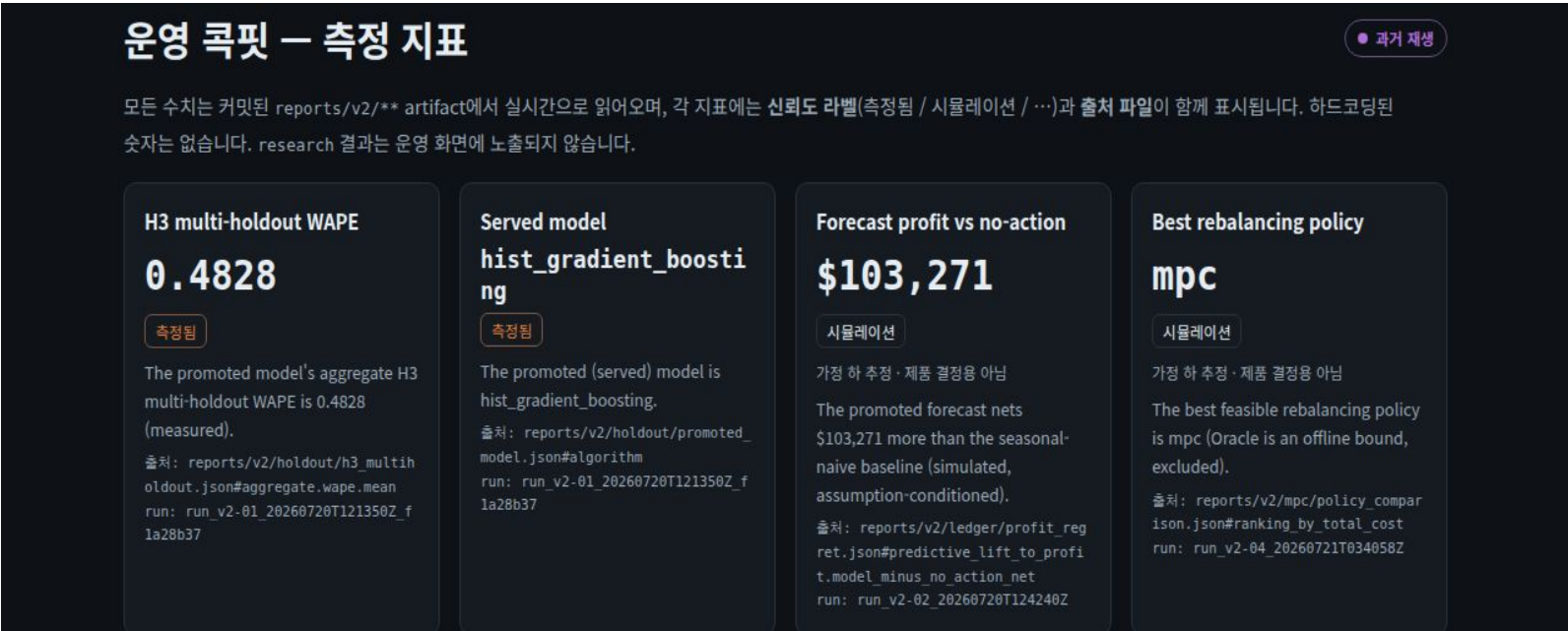
- **Neo4j graph DB:** (5,770 nodes, 11,850 edges)
- **FAISS:** search & dedup

4. Forecasting

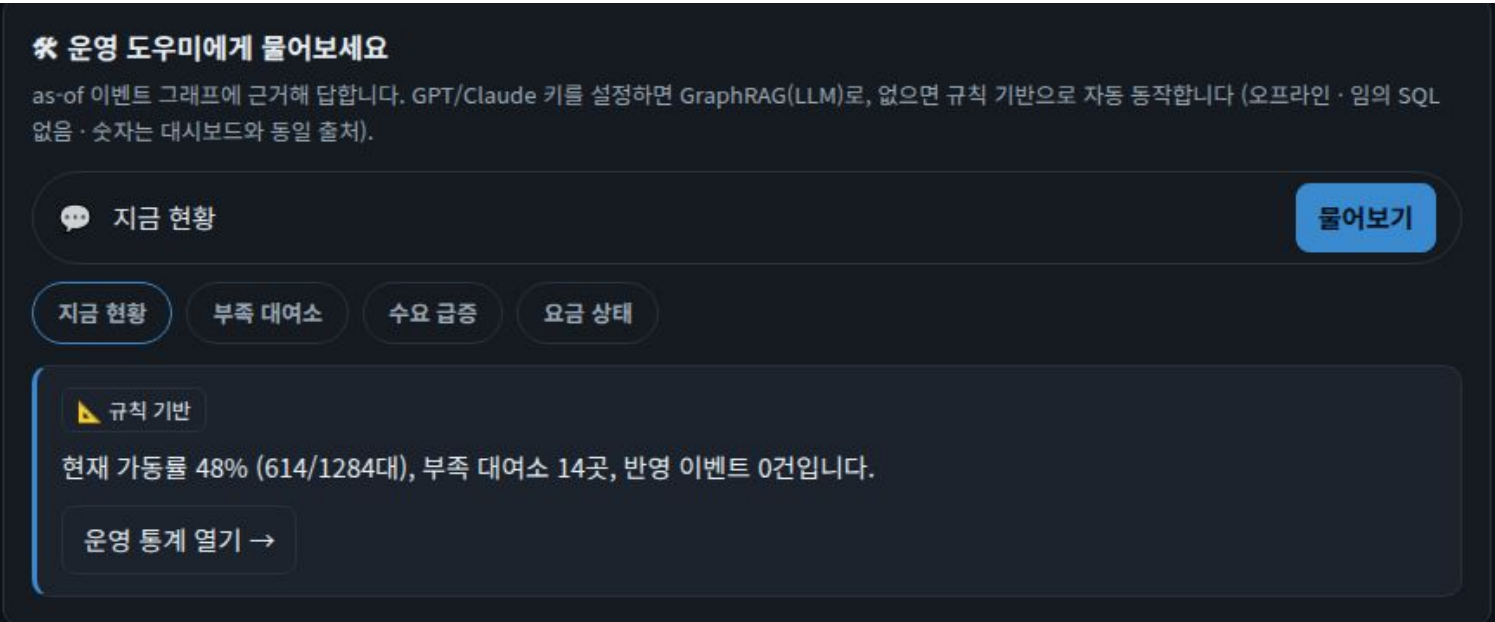
- GBM demand forecasting Train 14,759 / test 3,366
- Ablation: event features on/off

5. Optimization

MILP rebalancing with real constraints; pressure pricing; RL and QUBO experiments



Ops cockpit: metrics shown with confidence labels and sources



Ops copilot: grounded answers; numbers computed by tool calls

-1.65%

Relative WAPE improvement

Only pre-announced, well-located events helped

1. Structured city events catch demand drops

Validation: train Jan–May (14,759), test June (3,366)

Leakage: point-in-time joins, pre-forecast data only

Effect: beats baseline in 95.2% of 1,193 demand-drop windows

Stats: WAPE +1.65% (0.1013 → 0.0996), 95% CI [0.36, 5.11]%

Insight: only pre-announced, well-located events helped; vague news hurt

-73%

Shortage reduction

MPC optimization vs. no action

2. Forecasts feed rebalancing optimization

Baseline: greedy nearest-station moves

Optimization: MILP (HiGHS), extended to multi-period MPC

Effect: shortages –73% vs. no action; cost –36% vs. greedy

Optimality: matches exhaustive search on small instances

Policy: MPC beats RL; regret QL 247.8, PPO 202.9, MPC 21.6

0

Hallucinated citations

GraphRAG citation audit (10 questions)

3. Answers verified back to source text

Retrieval: GraphRAG links events to sources

Scoring: RAGAS faithfulness 1.0, relevancy 0.985, routing 1.0

Effect: hallucinations 10 → 0; correct answers 40% → 100%

Safeguards: block unsourced citations; rule-based fallback

한정연

JungYun Han, AI Engineer

At Samsung SDS I built and operated AI agents, ML and NLP models, and data pipelines. In a personal project I validated demand forecasting and optimization end to end.

CAREER

Samsung SDS, Sr. Data Consultant (2022–)

EDUCATION

2022 Ph.D. Theoretical Physics, UST/IBS

2018 M.Sc. Physics, Seoul Natl. Univ.

2015 B.Sc. Physics, Yonsei Univ.

CERTIFICATION

Big Data Engineering, Information Processing
OPIc IH, OPI IH

4 SCI papers, quantum optics/information
Patent filed (KR 2025-0139679)

SKILLS

Language	Python (PyTorch, scikit-learn, pandas)
Database	SQL (HiveQL, PostgreSQL), Vector DB (FAISS), Graph DB (Neo4j)
Language Model	Embedding Model, CrossEncoder, BERT, Sentence BERT, Longformer, Full fine-tuning, PEFT
ML / Optimization	Regression, classification, statistical testing, ensembles (XGBoost, LightGBM), time-series forecasting, MILP-based MPC
Generative AI	RAG, AI Agent, Knowledge Graph, GraphRAG
Infrastructure	Samsung Cloud Platform, air-gapped on-premise, Docker, GitLab CI, cron